

GATE CSE 2026

Predicted Paper

GATE CS 2026 - Predicted Paper 2: Systems and Applications

Focus: Operating Systems, DBMS, Computer Networks, Software Engineering

Total Questions: 65 | Total Marks: 100 | Duration: 180 min

Generated by PadhaiShuru | padhaishuru.com

General Aptitude

Q1. [1 mark] Choose OPPOSITE of: VINDICTIVE

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q2. [1 mark] Work rates: A in 10 days, B in 15 days. Together?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q3. [2 marks] Correct sentence:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q4. [1 mark] $\log_3(81) = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q5. [2 marks] $P(A)=0.3$, $P(B)=0.5$, independent. $P(A \cup B) = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q6. [1 mark] To bite the dust means:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q7. [1 mark] Boys:Girls = 3:2. Boys = 45. Total students?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q8. [2 marks] Synonym of: OBSTINATE

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q9. [2 marks] CI on Rs 5000 at 12% for 2 years = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q10. [1 mark] Find error: Each of the boys have completed their work.

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Engineering Mathematics

Q11. [1 mark] Rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Rows are proportional $\rightarrow$ rank 1.

Q12. [1 mark] Expected value of fair die roll?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $E[X] = (1+2+3+4+5+6)/6 = 3.5$.

Q13. [1 mark] integral from 0 to 1 of $x^2 dx$ = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Integral of x^2 is $x^3/3$. Evaluate at 1: $1/3$.

Q14. [2 marks] Function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x)=x^2$ is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Not one-one: $f(1)=f(-1)$. Not onto: negative values not in range.

Q15. [2 marks] A is 3×3 , $\det(A)=5$. $\det(A^{-1})$ = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $\det(A^{-1}) = 1/\det(A) = 1/5$.

Technical (CSE)

Q16. [1 mark] FCFS main drawback?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: FCFS causes convoy effect: short jobs wait behind long ones.

Q17. [2 marks] Average turnaround time (FCFS) for P1(20), P2(5), P3(2)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: FCFS order P1,P2,P3: CT=20,25,27. TAT=20,20,22. Average = $62/3 = 20.7$. Adjusted for realistic exam values.

Q18. [2 marks] Semaphore with initial value 1 is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Initial value 1 makes it binary semaphore. Used as mutex for mutual exclusion.

Q19. [2 marks] Bankers algorithm is for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Banker simulates allocation to ensure safe state -> deadlock avoidance.

Q20. [1 mark] Page replacement FIFO may show:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Belady's anomaly: more frames can increase page faults in FIFO.

Q21. [2 marks] EAT: TLB hit=80%, TLB=20ns, MM=100ns?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $EAT = 0.8 \cdot 20 + 0.2 \cdot (20 + 100) = 16 + 24 = 40ns$.

Q22. [1 mark] UNIX file system uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: UNIX/BSD use inode-based file systems.

Q23. [1 mark] fork() creates:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: fork() creates new child process with copy of parent address space.

Q24. [2 marks] LRU implemented using:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: LRU typically uses counters or timestamps per page to track recent use.

Q25. [1 mark] Necessary condition NOT required for deadlock?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Preemption actually prevents deadlock. Not a necessary condition.

Q26. [2 marks] Round robin with very large time quantum behaves like?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Very large quantum makes RR behave like FCFS.

Q27. [2 marks] LRU vs FIFO cache replacement:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: LRU generally better but depends on workload. Beladys anomaly affects FIFO.

Q28. [1 mark] BCNF eliminates:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BCNF is stricter than 3NF and eliminates all update anomalies.

Q29. [2 marks] Find names with salary > average salary?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Subquery in WHERE clause compares each salary to overall average.

Q30. [2 marks] 2NF violation example:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: 2NF requires no partial dependency on partial key.

Q31. [2 marks] ACID property for crash recovery?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Durability ensures committed changes survive crashes (write-ahead logging).

Q32. [1 mark] SELECT operation in RA is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: sigma = selection/conditional rows. pi = projection/columns.

Q33. [2 marks] B+ tree internal nodes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: B+ tree internal nodes store keys and child pointers. Data pointers only in leaves.

Q34. [1 mark] Weak entity has:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Weak entity has partial key (discriminator) + identifying relationship.

Q35. [2 marks] 3NF vs BCNF:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BCNF requires $X \rightarrow Y$ where X is superkey. 3NF allows $X \rightarrow Y$ if Y is prime. BCNF subset of 3NF.

Q36. [2 marks] Two-phase locking guarantees:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: 2PL ensures conflict-serializable schedules. Does NOT prevent deadlocks.

Q37. [1 mark] UNIQUE constraint allows:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: UNIQUE allows multiple NULLs (NULL != NULL in SQL). PRIMARY KEY does NOT allow NULLs.

Q38. [1 mark] CIDR notation /24 -> host bits?

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: /24 means 24 network bits. $32 - 24 = 8$ host bits.

Q39. [2 marks] Subnet mask for /20?

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: /20 -> 255.255.240.0 (20 network bits: $8+8+4=20$).

Q40. [2 marks] TCP 3-way handshake sequence:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: Standard: SYN -> SYN-ACK -> ACK.

Q41. [2 marks] TCP ssthresh after timeout?

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: On timeout: $ssthresh = cwnd/2$, $cwnd = 1$, restart slow start.

Q42. [1 mark] ICMP operates at:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: ICMP is network layer (IP protocol 1). Used for ping, traceroute.

Q43. [2 marks] CSMA/CD collision detected when?

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: CSMA/CD detects collisions while transmitting (jam signal sent).

Q44. [1 mark] DNS uses:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: DNS uses UDP for queries, TCP for zone transfers.

Q45. [2 marks] RIP uses:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: RIP = distance vector. OSPF = link state. BGP = path vector.

Q46. [1 mark] Testing individual modules:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: Unit testing = individual module/function testing.

Q47. [2 marks] Testing interactions between modules:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: Integration testing tests module interactions.

Q48. [1 mark] Waterfall model phases order:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: Waterfall: Requirements -> Design -> Implementation -> Testing -> Maintenance.

Q49. [2 marks] COCOMO estimates:

- @. [object Object]
A. [object Object]

- B. [object Object]
- C. [object Object]

Explanation: COCOMO estimates effort (person-months) and schedule (months).

Q50. [1 mark] SRS stands for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: SRS = Software Requirements Specification.

Q51. [2 marks] Equivalence partitioning is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Equivalence partitioning divides input domain -> black-box technique.

Q52. [2 marks] Cyclomatic complexity = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: McCabe: $V(G) = E - N + 2P$ where E =edges, N =nodes, P =components.

Q53. [1 mark] Scrum is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Scrum is an Agile framework with sprints, daily standups, retrospectives.

Q54. [1 mark] Which is block-level element?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: div is block-level. span, a, img are inline.

Q55. [1 mark] CSS specificity of #id .class?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $ID=100(0,1,0,0) + Class=10(0,0,1,0) = 110 \rightarrow (0,1,1,0)$.

Q56. [1 mark] HTTP 404 means:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: 404 = Not Found. 200=OK, 500=Server Error, 401=Unauthorized.

Q57. [2 marks] typeof null in JS?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: typeof null returns 'object' - a well-known JS quirk.

Q58. [2 marks] HTTP method for updating existing resource?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: PUT updates existing resource. POST creates. DELETE removes.

Q59. [1 mark] IP address class B first octet range:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Class B: 128-191. Class A: 0-127, C: 192-223, D: 224-239 (multicast).

Q60. [2 marks] Subnets from /24 using /26 mask?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: /26 from /24: $2^{(26-24)} = 2^2 = 4$ subnets.

Q61. [2 marks] TCP congestion window on timeout:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: On timeout: $ssthresh=cwnd/2$, $cwnd=1$, restart slow start.

Q62. [1 mark] SMTP is for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: SMTP = Simple Mail Transfer Protocol for email.

Q63. [2 marks] CRC polynomial degree 3. Message 1010. CRC bits?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Generator degree 3 -> CRC = 3 bits appended to message.

Q64. [1 mark] SJF is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: SJF can be non-preemptive or preemptive (SRTF = Shortest Remaining Time First).

Q65. [2 marks] Page fault service time (no TLB, page not in memory)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Page fault: fetch page from disk -> load into memory -> restart instruction. Dominated by disk I/O.

Answer Key

General Aptitude

Q1: B
Q2: A
Q3: B
Q4: C
Q5: B
Q6: A
Q7: B
Q8: B
Q9: A
Q10: C

Engineering Mathematics

Q11: B
Q12: B
Q13: B
Q14: D
Q15: B

Technical (CSE)

Q16: B
Q17: C
Q18: D
Q19: B
Q20: B
Q21: A
Q22: B
Q23: B
Q24: C
Q25: C
Q26: A
Q27: C
Q28: C
Q29: C
Q30: A
Q31: D
Q32: B
Q33: C
Q34: B
Q35: A
Q36: B
Q37: A
Q38: A
Q39: A
Q40: A
Q41: A
Q42: C
Q43: B
Q44: C
Q45: B
Q46: B
Q47: B
Q48: B
Q49: C
Q50: A
Q51: A
Q52: A
Q53: B
Q54: B
Q55: B
Q56: B
Q57: B
Q58: C
Q59: B
Q60: B
Q61: B
Q62: B

Q63: C
Q64: B
Q65: B